

Assignment 3

Xiaochen Liu

2025-08-3

Data import and cleaning

The data used in this assignment is from the real estate market in Taiwan, which includes various features of houses such as house age, distance to the nearest MRT station, and house price. The dataset is first cleaned to focus on houses that are less than 20 years old.

Data exploration

At the beginning, the correlation matrix is computed to understand the relationships between these variables. The correlation matrix shows the pairwise correlations between the numeric variables in the dataset. The values range from -1 to 1, where -1 indicates a perfect negative correlation, 0 indicates no correlation, and 1 indicates a perfect positive correlation.

Since we aims to conclude the variable that has the most significant impact on house price, we will focus on the correlation between house prices and other variables. The most significant correlations are found between house prices and MRT distance, which is -0.7, indicating that as the distance to MRT increases, the price tends to decrease. The number of convenience stores has a positive correlation with house price, suggesting that houses have more convenience stores nearby tend to be more expensive.

```
##           transaction_date  house_age mrt_distance convenience_stores
## transaction_date          1.00000000  0.08374664   0.04786342    0.03214026
## house_age                 0.08374664  1.00000000   0.30353643   -0.25702245
## mrt_distance              0.04786342  0.30353643   1.00000000   -0.61758560
## convenience_stores        0.03214026 -0.25702245  -0.61758560    1.00000000
## latitude                  0.05795883 -0.10938295  -0.57060366    0.39474106
## longitude                 -0.02949622 -0.21137032  -0.73453243    0.41352327
## house_price               0.06384619 -0.45799337  -0.70647298    0.58849237
##           latitude  longitude house_price
## transaction_date  0.05795883 -0.02949622  0.06384619
## house_age        -0.10938295 -0.21137032 -0.45799337
## mrt_distance     -0.57060366 -0.73453243 -0.70647298
## convenience_stores 0.39474106  0.41352327  0.58849237
## latitude          1.00000000  0.33990823  0.53873949
## longitude          0.33990823  1.00000000  0.49027501
## house_price        0.53873949  0.49027501  1.00000000
```

```
print(plot1)
```

